6.S974 Games, Learning, and Security
HW2

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Problem 1 (Cybersecurity with Incomplete Information)

(a) **Pure strategies.** A strategy specifies a choice for each type.

- Attacker ($\theta_1 = 4, \theta_1 = 8$): $\{(A, A), (A, N), (N, A), (N, N)\}$
- Company: ($\theta_2 = 1, \theta_2 = 3$): $\{(P, P), (P, U), (U, P), (U, U)\}$

A strategy profile is a tuple of one pure strategy per player. In this case each player has four pure strategies, so there are $4 \times 4 = 16$ strategy profiles.

(b) **Prior and beliefs.** Types are equally likely to occur and independent so

$$p(\theta_1, \theta_2) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4}, \qquad \forall\, \theta_1,\ \forall\, \theta_2$$

Beliefs are conditional on the attacker or the company's types

$$\mu_1(\theta_2 = 1 \mid \theta_1 = 4) = \frac{p(4,1)}{p(4,1) + p(4,3)} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \tfrac{1}{2}$$

Because the prior is uniform over (θ_1, θ_2), this belief holds for every player's type for every other player's type so

$$\mu_1(\theta_2 \mid \theta_1) = \mu_2(\theta_1 \mid \theta_2) = \tfrac{1}{2}$$

(c) **Pure strategy best response.** Expected payoff when company plays P is $E_{A\sim\alpha}[u_2(P, A)] = -\alpha\theta_2 - (1-\alpha)\theta_2 = -\theta_2$ and when company plays U is $E_{A\sim\alpha}[u_2(U, A)] = -4\alpha$, so

$$BR_2(\alpha) = \begin{cases} P & \theta_2 < 4\alpha \\ \{P, U\} & \theta_2 = 4\alpha \\ U & \text{else} \end{cases} \implies \begin{cases} (P,P) & \alpha \geq \frac{3}{4} \\ (P,U) & \frac{1}{4} \leq \alpha \leq \frac{3}{4} \\ (U,P) & \text{never} \\ (U,U) & \alpha \leq \frac{1}{4} \end{cases}$$

And for the attacker we get $E_{P\sim\pi}[u_1(P, A)] = -\pi + (1-\pi)(\theta_1 - 1) = (1-\pi)\theta_1 - 1$ and $E_{P\sim\pi}[u_1(P, N)] = 0$

$$BR_1(\pi) = \begin{cases} A & \pi < 1 - \frac{1}{\theta_1} \\ \{A, N\} & \pi = 1 - \frac{1}{\theta_1} \\ N & \text{else} \end{cases} \implies \begin{cases} (A,A) & \pi \leq \frac{3}{4} \\ (A,N) & \text{never} \\ (N,A) & \frac{3}{4} \leq \pi \leq \frac{7}{8} \\ (N,N) & \pi \geq \frac{7}{8} \end{cases}$$

(d) **Does a pure strategy Bayes Nash Equilibrium exist?**

- Begin with attacker pure strategy (A, A): $\sigma_1(A \mid \theta_1 = 4) = 1 = \sigma_1(A \mid \theta_1 = 8)$ so $\alpha = 1$. From part (c) $BR_2(1) = (P, P)$. By the same logic company pure strategy (P, P) yields $\pi = 1$ $BR_1(1) = (N, N)$. Repeating, attacker pure strategy (N, N) yields $\alpha = 0$ and $BR_2(0) = (U, U)$. Company pure strategy (U, U) yields $\pi = 0$ and $BR_1(0) = (A, A)$, returning us to the start of the cycle. So there's no pure strategy Bayes Nash Equilibrium with any combination of $((A, A), (N, N))$ and $((P, P), (U, U))$
- Take attacker pure strategy (A, N) yielding $\alpha = \frac{1}{2}$ and $BR_2(\frac{1}{2}) = (P, U)$. The company's best response yields $\pi = \frac{1}{2}$ and $BR_1(\frac{1}{2}) = (A, A)$, returning us to the first cycle.
- Take attacker strategy (N, A) yielding $\alpha = \frac{1}{2}$. This produces the same $BR_2(\frac{1}{2}) = (P, U)$ above returning us to the first cycle.

Hence by process of elimination, there is no pure strategy Bayes Nash Equilibrium.

(e) **Bayesian Nash Equilibrium in behavioral strategies.**

Necessary condition $\pi = \frac{3}{4}$ By contradiction suppose $\pi \neq \frac{3}{4}$ in equilibrium. Recall that a mixed action is a best response only if every action in its support is a pure best response by theorem.

Case 1 $(\pi > 3/4)$**.** For any $\pi > 3/4$ attacker type $\theta_1 = 4$ has only a pure best response N (because mixed best response requires pure best response in its support) and $\sigma_1(A \mid \theta_1 = 4) = 0$. So $\alpha = \sigma_1(A \mid \theta_1 = 8)/2$, which implies $\alpha \leq 1/2$.

For any $\alpha \leq 1/2$ the company type $\theta_2 = 3$ has only a pure best response U and $\sigma_2(P \mid \theta_2 = 3) = 0$. So $\pi = \sigma_2(P \mid \theta_2 = 1)/2$, which implies $\pi \leq 1/2$, a contradiction to $\pi > 3/4$.

Case 2 $(\pi < 3/4)$ For any $\pi < 3/4$ the attacker has only a pure best response (A, A) and $\alpha = 1$. For $\alpha = 1$ the company has only a pure best response (P, P) and $\pi = 1$, contradicting $\pi < 3/4$.

Therefore if there exists a Bayesian Nash Equilibrium a necessary condition would be $\pi = 3/4$.

Mixed strategy. The probability of protecting $\pi = 3/4$ gives $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) = 3/2$, which implies at least one company type randomizes, since pure company strategies give $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) \in \{0, 1, 2\}$.

Moreover, for $\pi = 3/4$ the best response of attacker type $\theta_1 = 8$ is A, implying a probability of attack $\alpha \geq 1/2$.

For the company to have a mixed strategy and $\alpha \geq 1/2$, it must be that $\alpha = 3/4$ so that the mixed strategy best response can have a support greater than 1: the company type $\theta_1 = 3$ is indifferent at $\alpha = 3/4$ and type $\theta_1 = 1$ is not.

This support tells us that company type $\theta_2 = 1$ has a pure strategy best response of P and $\sigma_2(P \mid 1) = 1$. Plugging this into our equation for π above we get $\sigma_2(P \mid 3) = 1/2$.

Note that $\sigma_1(A \mid 8) = 1$ follows directly from attacker type $\theta_1 = 8$'s best response of A. Solving $\alpha = 3/4 = (1 + \sigma_1(A \mid 4))/2$ gives $\sigma_1(A \mid 4) = 1/2$.

Therefore if there is a mixed strategy Bayesian Nash Equilibrium it must be that

$$\sigma_2(P \mid 3) = 1/2 = \sigma_1(A \mid 4), \quad \sigma_2(P \mid 1) = 1 = \sigma_1(A \mid 8).$$

Suffice to validate.

Validation. From definition, the strategy profile is a Bayes Nash Equilibrium if for every player and every type the strategy is a best response. At the candidate profile $\pi = 3/4$ and $\alpha = 3/4$.

$$\begin{aligned}
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 4)) = \{A, N\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 4) = \{A, N\} \\
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 8)) = \{A\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 8) = \{A\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 1)) = \{P\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 1) = \{P\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 3)) = \{P, U\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 3) = \{P, U\}
\end{aligned}$$

Each condition above was necessary for equilibrium, so it is the unique BNE.

Problem 2 (Signaling through Safety Certification)

(a) **Game tree.**

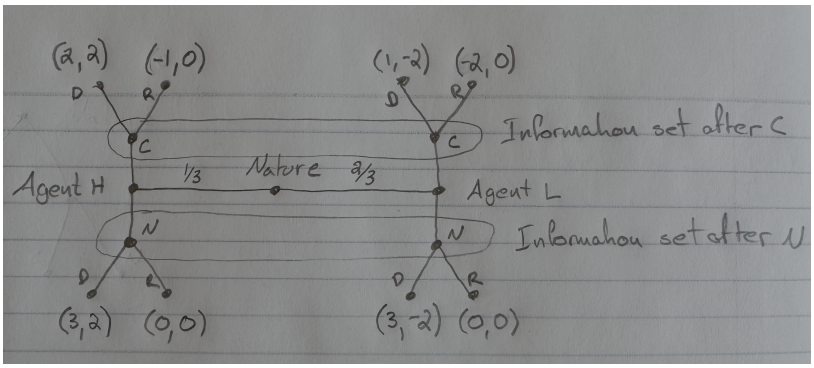


Figure 1: Problem 2 Part (a) game tree

(b) **Does a separating PBE exist?** A separating PBE exists if a strategy profile (s^*, μ^*) satisfies

1. *Sequential rationality.* At every information set I each player's continuation strategy $s_i^*\big|_I$ maximizes expected payoff given beliefs $\mu_i^*(\cdot \mid I)$ and the other players' continuation strategies $s_{-i}^*\big|_I$.
2. *Belief consistency.* At information sets I reached with positive probability, beliefs $\mu_i^*(\cdot \mid I)$ are obtained from the prior and equilibrium strategies s^* using Bayes' rule.
3. *Separating.* Each provider type chooses a different signal.

Candidate provider strategy profiles are $s_1(H, L) \in \{(C, N), (N, C)\}$.

Profile (C, N)**.** Under belief consistency the operator's beliefs $\mu_2(\cdot \mid m_1)$ are

$$\mu_2(H \mid C) = \frac{P(C \mid H)P(H)}{P(C \mid H)P(H) + P(C \mid L)P(L)} = \frac{1 * 1/3}{1 * 1/3 + 0} = 1, \qquad \mu_2(L \mid C) = 0,$$
$$\mu_2(H \mid N) = 0, \qquad \mu_2(L \mid N) = 1.$$

Under sequential rationality, the operator's strategy is $s_2(C) = D$ and $s_2(N) = R$.

In equilibrium, the provider should not have an incentive to deviate. When $\theta_1 = H$, selecting N decreases provider payoff from $2 \to 0$. When $\theta_1 = L$, selecting C increases provider payoff from $0 \to 1$. So (C, N) is not a separating PBE.

Profile (N, C)**.** Operator beliefs are

$$\mu_2(H \mid C) = 0, \qquad \mu_2(L \mid C) = 1, \qquad \mu_2(H \mid N) = 1, \qquad \mu_2(L \mid N) = 0.$$

The operator's strategy is $s_2(C) = R$ and $s_2(N) = D$. When $\theta_1 = H$, selecting C decreases provider payoff from $3 \to -1$. When $\theta_1 = L$, selecting N increases provider payoff from $-2 \to 3$. So (N, C) is not a separating PBE.

Therefore, there does not exist a separating PBE.

(c) **Pooling PBE.** Now consider a pooling PBE where $s_1(H) = s_1(L) = N$.

The operator's beliefs are

$$\mu_2(H \mid N) = \frac{P(N \mid H)P(H)}{P(N \mid H)P(H) + P(N \mid L)P(L)} = \frac{1 * 1/3}{1 * 1/3 + 1 * 2/3} = 1/3, \quad \mu_2(L \mid N) = 2/3.$$

The operator's strategy is based on expected payoffs

$$E_{\theta_1}[u_2(N, R \mid \theta_1)] = 0, \qquad E_{\theta_1}[u_2(N, D \mid \theta_1)] = 1/3 * 2 + 2/3 * (-2) = -2/3$$

and its strategy is $s_2(N) = R$.

Bayes rule cannot determine $\mu_2(\cdot \mid C)$. Let $\mu_2 := \mu_2(H \mid C)$ and ask what beliefs can discourage both provider types from deviating from N:

$$E_{\theta_1}[u_2(C, R \mid \theta_1)] = 0, \qquad E_{\theta_1}[u_2(C, D \mid \theta_1)] = \mu_2 * 2 + (1 - \mu_2) * (-2) = 4\mu_2 - 2.$$

Comparing the two expected payoffs we see for any $\mu_2 \leq 1/2$ the operator's best response would be $s_2(C) = R$. Choose $\mu_2 = 0$ so $s_2(N) = s_2(C) = R$ and examine the provider's best response.

When $\theta_1 = H$, selecting C decreases payoffs from $0 \to -1$ and when $\theta_1 = L$, selecting C decreases payoff from $0 \to -2$.

Therefore the profile $s^* = (s_1(H) = s_1(L) = N, s_2(C) = s_2(N) = R)$, together with beliefs $\mu_2(H \mid N) = 1/3$ (from Bayes' rule) and $\mu_2(H \mid C) = 0$ (chosen), is a pooling PBE.

(d) **Behavioral strategy for a semi-separating equilibrium.**

Suppose $\sigma_1(C \mid H) = 1,\ \sigma_1(C \mid L) = p \in (0, 1)$.

Operator beliefs.

$$\mu_2(H \mid C) = \frac{\sigma_1(C \mid H)P(H)}{\sigma_1(C \mid H)P(H) + pP(L)} = \frac{1 * 1/3}{1 * 1/3 + 2p/3} = \frac{1}{1 + 2p}, \qquad \mu_2(L \mid C) = \frac{2p}{1 + 2p}$$
$$\mu_2(H \mid N) = 0, \qquad \mu_2(L \mid N) = 1.$$

$\sigma_2(\cdot \mid \mathbf{N})$. When the provider plays N the operator has a pure best response of $s_2(N) = R$ since

$$E_{\theta_1}[u_2(N, D \mid \theta_1)] = 0 * 2 + 1 * (-2) < 0 = E_{\theta_1}[u_2(N, R \mid \theta_1)]$$

So $\sigma_2(R \mid N) = 1 = 1 - \sigma_2(D \mid N)$.

$\sigma_1(\cdot \mid \mathbf{L})$. When the provider plays C, the operator can play a mixed strategy only if both pure strategies are in its best response. This occurs when

$$\begin{aligned} E_{\theta_1 \sim \mu_2}[u_2(C, D \mid \theta_1)] &= 0 = E_{\theta_1 \sim \mu_2}[u_2(C, R \mid \theta_1)] \\ 2 * \mu_2(H \mid C) + -2 * \mu_2(L \mid C) &= 0 \\ 2/(1 + 2p) - 4p/(1 + 2p) &= 0 \\ p &= 1/2 \end{aligned}$$

So $\sigma_1(C \mid L) = 1/2 = \sigma_1(N \mid L)$.

$\sigma_2(\cdot \mid \mathbf{C})$. Let $\sigma_2(D \mid C) = q$ and recall $\sigma_2(R \mid N) = 1$. For a behavioral strategy equilibrium we must find q such that the provider of type $\theta_1 = L$ is indifferent between actions C and N:

$$\begin{aligned} E_{\sigma_2}[u_1(C, \sigma_2 \mid L)] &= E_{\sigma_2}[u_1(N, \sigma_2 \mid L)] \\ q * 1 + (1-q)(-2) &= 0 = u_1(N, R \mid L) \\ q &= 2/3 \end{aligned}$$

So $\sigma_2(D \mid C) = 2/3 = 1 - \sigma_2(R \mid C)$.

$\sigma_1(\cdot \mid \mathbf{H})$ It suffices to show $\sigma_1(C \mid H) = 1 = 1 - \sigma_1(N \mid H)$ is a best response:

$$E_{\sigma_2}[u_1(C, \sigma_2 \mid H)] = 2/3 * 2 + 1/3(-1) = 1 \geq 0 = E_{\sigma_2}[u_1(N, \sigma_2 \mid H)]$$

So $s_1(H) = C$ is a pure strategy best response confirming the check.

Equilibrium. Therefore the following strategy profile

$$\begin{aligned} &\sigma_1(C \mid H) = 1, \quad \sigma_1(N \mid H) = 0, \quad \sigma_1(C \mid L) = 1/2, \quad \sigma_1(N \mid L) = 1/2. \\ &\sigma_2(D \mid C) = 2/3, \quad \sigma_2(R \mid C) = 1/3, \quad \sigma_2(D \mid N) = 0, \quad \sigma_2(R \mid N) = 1. \end{aligned}$$

with beliefs

$$\begin{aligned} \mu_2(H \mid C) &= 1/2, & \mu_2(L \mid C) &= 1/2 \\ \mu_2(H \mid N) &= 0, & \mu_2(L \mid N) &= 1. \end{aligned}$$

is a PBE in behavioral strategies.

Probability of unreliable and deployed.

$$\begin{aligned} P(D, L) &= P(D \mid L)P(L) \\ &= \big[P(D \mid L, C)P(C \mid L) + P(D \mid L, N)P(N \mid L)\big]P(L) && \text{(law of total probability)} \\ &= \big[\sigma_2(D \mid C)\sigma_1(C \mid L) + \sigma_2(D \mid N)\sigma_1(N \mid L)\big]P(L) && \text{(plugging in strategy profiles)} \\ &= \big[\tfrac{2}{3} \cdot \tfrac{1}{2} + 0 \cdot \tfrac{1}{2}\big]\tfrac{2}{3} \\ &= \tfrac{2}{9} \end{aligned}$$

Problem 3 (Traffic routing problem)

(a) **Wardrop equilibrium.** For signal s all paths that carry positive flow must have same expected latency:

$$x^W(\mu) = \begin{cases} 0 & \mu \in [0, \frac{1}{3}] \quad \big(\mu \leq \frac{1}{3} + 2 * 0\big) \\ \frac{\mu - 1/3}{2} & \mu \in [\frac{1}{3}, 1] \\ 1 & \mu \geq \frac{1}{3} + 2 = \frac{7}{3} \ \text{ (not possible)} \end{cases}$$

Conditional expected total latency for $\mu \in [\frac{1}{3}, 1]$ is

$$\begin{aligned} C(\mu) &= \left(1 - \frac{\mu - 1/3}{2}\right)\mu + \left(\frac{\mu - 1/3}{2}\right)\left(\tfrac{1}{3} + 2\frac{\mu - 1/3}{2}\right) \\ &= \mu - \frac{\mu^2 - \mu/3}{2} + \frac{\mu^2 - \mu/3}{2} = \mu \end{aligned}$$

For $\mu \in [0, \frac{1}{3}]$ it's

$$C(\mu) = (1 - 0)\mu + 0(\tfrac{1}{3} + 2 * 0) = \mu$$

So $C(\mu) = \mu$, $\forall \mu \in [0, 1]$.

Uniqueness. Fix μ and suppose the Wardrop equilibrium is not unique. Then there exists $0 \le x < x' \le 1$ such that both flows satisfy the Wardrop condition.
$0 < x' \implies$ path 2 latency is no worse than path 1 latency: $\mu \ge \frac{1}{3} + 2x'$.
$x < 1 \implies$ path 1 latency is no worse than path 2 latency: $\mu \le \frac{1}{3} + 2x$. So

$$\tfrac{1}{3} + 2x' \le \mu \le \tfrac{1}{3} + 2x \implies x' \le \frac{\mu - \frac{1}{3}}{2} \le x$$

which contradicts $x < x'$. Therefore the equilibrium is unique.

(b) **Mean-posterior identity.**

$$\begin{aligned}
\sum_{\{s \in S : \lambda_s > 0\}} \lambda_s \mu_s &= \sum_{\{s \in S : P(s) > 0\}} P(s) P(\theta = 1 \mid s) && \text{(definition)} \\
&= \sum_{\{s \in S : P(s) > 0\}} P(\theta = 1, s) = \sum_{\{s \in S : P(s) > 0\}} P(s \mid \theta = 1) P(\theta = 1) && \text{(Bayes)} \\
&= p \sum_{\{s \in S : P(s) > 0\}} \pi(s \mid \theta = 1) && \text{(law of total probability)} \\
&= p \cdot 1
\end{aligned}$$

Where the final step holds for $P(s) > 0$ since for $p \in (0, 1)$ and every $P(s) = 0$,

$$0 = P(s) = p\pi(s \mid \theta = 1) + (1 - p)\pi(s \mid \theta = 0) \implies \pi(s \mid \theta = 1) = 0$$

(c) **Public signal rule.**

$$\begin{aligned}
C(\mu_s) &= \sum_{\theta} P(\theta \mid s) \cdot C(x^W(\mu_s), \theta) = \mu_s && \text{(conditional expected cost from part (a))} \\
E_s[C(\mu_s)] &= \sum_{s} P(s) C(\mu_s) && \text{(ex ante expected cost)} \\
&= \sum_{s} \lambda_s \mu_s && \text{(from part (b))} \\
&= p && \text{(from part (c))}
\end{aligned}$$

So ex ante expected total latency does not depend on the public signal rule $\pi(s \mid \theta)$ hence any public signal rule gives the same value.

Full revelation. For $s \in \{0, 1\}$, $\pi(s \mid \theta) = 1$ for $s = \theta$ and 0 otherwise. Then

$$\mu(\theta = 1 \mid s) = \frac{\pi(s \mid \theta = 1)p}{\pi(s \mid \theta = 1)p + 0} = 1$$
$$\mu(\theta = 0 \mid s) = \frac{\pi(s \mid \theta = 0)(1 - p)}{\pi(s \mid \theta = 0)(1 - p) + 0} = 1$$

From part (a), $x^W(\mu_s = 1) = (1 - \frac{1}{3})/2 = \frac{1}{3}$.

No information. For $s \in \{0\}$, $\pi(s \mid \theta) = 1$ for $\theta \in \{0, 1\}$. Then

$$\mu(\theta = 1 \mid s) = \frac{\pi(s \mid \theta = 1)p}{\pi(s \mid \theta = 1)p + \pi(s \mid \theta = 0)(1 - p)} = p$$

From part (a),

$$x^W(\mu_s = p) = \begin{cases} 0 & p \in [0, \frac{1}{3}] \\ \frac{p - 1/3}{2} & p \in [\frac{1}{3}, 1] \end{cases}$$

(d) **Private information general rule.** Recommendation probabilities are

$$P(r=2) = P(r=2 \mid \theta=1)P(\theta=1) + P(r=2 \mid \theta=0)P(\theta=0) = q_1 p + q_0(1-p)$$
$$P(r=1) = P(r=1 \mid \theta=1)P(\theta=1) + P(r=1 \mid \theta=0)P(\theta=0) = (1-q_1)p + (1-q_0)(1-p)$$

The posterior probabilities of $\theta = 1$ are

$$P(\theta=1 \mid r=2) = \frac{P(r=2 \mid \theta=1)p}{P(r=2)} = \frac{q_1 p}{q_1 p + q_0(1-p)}$$
$$P(\theta=1 \mid r=1) = \frac{P(r=1 \mid \theta=1)p}{P(r=1)} = \frac{(1-q_1)p}{(1-q_1)p + (1-q_0)(1-p)}$$

Recommendations are incentive compatible for a driver if following the recommendation has expected latency at most as high as deviating. Written below without conditional probability denominators (which cancel as shown in $r=2$)

$$\begin{aligned}
r=2: \qquad & E_{\theta|r=2}[\tfrac{1}{3} + 2q_\theta] \le E_{\theta|r=2}[\theta] \\
& (\tfrac{1}{3} + 2q_1)P(\theta=1 \mid r=2) + (\tfrac{1}{3} + 2q_0)(P(\theta=0 \mid r=2)) \le P(\theta=1 \mid r=2) \\
& \left(\tfrac{1}{3} + 2q_1\right)\frac{q_1 p}{q_1 p + q_0(1-p)} + \left(\tfrac{1}{3} + 2q_0\right)\frac{q_0(1-p)}{q_1 p + q_0(1-p)} \le \frac{q_1 p}{q_1 p + q_0(1-p)} \\
& (\tfrac{1}{3} + 2q_1)q_1 p + (\tfrac{1}{3} + 2q_0)(1-p)q_0 \le q_1 p
\end{aligned}$$

$$\begin{aligned}
r=1: \qquad & E_{\theta|r=1}[\tfrac{1}{3} + 2q_\theta] \ge E_{\theta|r=1}[\theta] \\
& (\tfrac{1}{3} + 2q_1)P(\theta=1 \mid r=1) + (\tfrac{1}{3} + 2q_0)P(\theta=0 \mid r=1) \ge P(\theta=1 \mid r=1) \\
& (\tfrac{1}{3} + 2q_1)(1-q_1)p + (\tfrac{1}{3} + 2q_0)(1-q_0)(1-p) \ge (1-q_1)p
\end{aligned}$$

Bayes rule places $P(r)$ in the denominator of the posterior expectation, so zero-probability recommendations have undefined posterior expectations of $\theta = 1$. These recommendations can safely be dropped from the set of recommendations as are all $s : \lambda_s = 0$ in part (c) since for $p \in (0,1)$ and $P(r) = 0$

$$0 = P(r) = pP(r \mid \theta=1) + (1-p)P(r \mid \theta=0) \implies P(r \mid \theta=1) = 0.$$

Expected platform costs is

$$\begin{aligned}
E_\theta\left[C(q_\theta, \theta)\right] = J(q_0, q_1) &= (1-p)(C(q_0, 0)) + p(C(q_1, 1)) \\
&= (1-p)\left[(1-q_0)0 + q_0\left(\tfrac{1}{3} + 2q_0\right)\right] + p\left[(1-q_1)1 + q_1\left(\tfrac{1}{3} + 2q_1\right)\right] \\
&= (1-p)\left[2q_0^2 + \tfrac{1}{3}q_0\right] + p\left[2q_1^2 - \tfrac{2}{3}q_1 + 1\right]
\end{aligned}$$

(e) **Socially optimal flow, costs, and IC compatibility.**

Socially optimal flow.

$\theta = 1$: $\quad C(x,1) = 1 - x + \frac{x}{3} + 2x^2 = 2x^2 - \frac{2x}{3} + 1$

$\frac{dC(x,1)}{dx} = 0 \implies 0 = 4x - \frac{2}{3} \implies x = \frac{1}{6}$ (This is a minimum since second derivative is positive)

$\theta = 0$: $\quad C(x,0) = \frac{x}{3} + 2x^2$

$\frac{dC(x,0)}{dx} = 0 \implies 0 = \frac{1}{3} + 4x \implies x = -\frac{1}{12}$ (no critical point on $(0,1)$ so minimum is on the boundary)

$C(0,0) = 0, \quad C(1,0) = \frac{1}{3} + 2 = \frac{7}{3}$ (so x=0 at the boundary achieves minimum)

So the optima are $(x_0^S, x_1^S) = (0, 1/6)$.

Ex ante expected cost.

$$E_\theta[C(x,\theta)] = p \cdot (\tfrac{5}{6} + \tfrac{1}{6}(\tfrac{1}{3} + \tfrac{2}{6})) + (1-p)0 = p(\tfrac{15}{18} + \tfrac{2}{18}) = \tfrac{17}{18}p$$

Private recommendation rule. Plugging in socially optimal flows $(q_0, q_1) = (0, 1/6)$ into the inequalities from part (d)

$$r = 2: \quad (1/3 + 1/3)(p/6) + (1/3 + 2 \cdot 0)(1-p) \cdot 0 \le p/6 \implies 2p/18 \le p/6 \implies 2p \le 3p$$
($r = 2$ is incentive compatible $\forall\, p \in (0,1)$)

$$r = 1: \quad (1/3 + 1/3)(5p/6) + (1/3 + 0)(1-p) \ge 5p/6 \implies 11p/18 \le 1/3 \implies p \le 6/11$$

so recommendations r are incentive compatible for $p \le 6/11$, determined by $r = 1$.

Compare to public information rules. We've just shown that for $p \le 6/11$, a private information rule can achieve the ex ante social optimum latency at $17p/18$, which for $p \in (0,1)$ is lower than the public rule ex ante social optimum latency from part (c) of p.

A private rule can give different drivers different posteriors, depending on which recommendation they receive. Knowing the (q_0, q_1) that achieve the social optimum, we see posteriors are

$$P(\theta = 1 \mid r = 2) = \tfrac{p}{6}/(\tfrac{p}{6} + 0) = 1,$$
$$P(\theta = 1 \mid r = 1) = \tfrac{5p}{6}/(\tfrac{5p}{6} + (1-p)).$$

Drivers told $r = 2$ know with certainty $\theta = 1$ and follow the recommendation. Drivers told $r = 1$ remain uncertain enough that the recommendation path looks no worse than the deviation. So with these posteriors on the recommendations, the planner can hold the socially optimal flows.

(f) **Cost minimization.**

$$\begin{aligned}
\min_{q_0,q_1} \quad & (1-p)\left[2q_0^2 + \tfrac{1}{3}q_0\right] + p\left[2q_1^2 - \tfrac{2}{3}q_1 + 1\right] && (J(q_0,q_1)) \\
\text{s.t.} \quad & (\tfrac{1}{3} + 2q_1)q_1 p + (\tfrac{1}{3} + 2q_0)(1-p)q_0 \le q_1 p && (r = 2 \text{ IC}) \\
& (\tfrac{1}{3} + 2q_1)(1-q_1)p + (\tfrac{1}{3} + 2q_0)(1-q_0)(1-p) \ge (1-q_1)p && (r = 1 \text{ IC}) \\
& (q_0, q_1) \in [0,1]^2
\end{aligned}$$

```
# %%
import numpy as np
import cvxpy as cp
import polars as pl

pl.Config.set_tbl_cols(-1)

# %%
# ======================
# Problem 3 Part (f)
# ======================
def private_recs(p):
    # variables
    q0 = cp.Variable()
    q1 = cp.Variable()
    # constraints
    const = [
        q0 >= 0,
        q0 <= 1,
        q1 >= 0,
        q1 <= 1,
    ]
    const += [
```

```
        (q1 / 3 + 2 * cp.square(q1)) * p + (q0 / 3 + 2 * cp.square(q0)) * (1 - p)
        <= q1 * p
    ]
    const += [
        (1 / 3 + 5 * q1 / 3 - 2 * cp.square(q1)) * p
        + (1 / 3 + 5 * q0 / 3 - 2 * cp.square(q0)) * (1 - p)
        >= (1 - q1) * p,
    ]
    # objective
    obj = (1 - p) * (2 * cp.square(q0) + q0 / 3) + p * (
        2 * cp.square(q1) - 2 * q1 / 3 + 1
    )
    # problem
    prob = cp.Problem(cp.Minimize(obj), const)
    prob.solve()
    return float(q0.value), float(q1.value), float(prob.value), prob.status


results = []
for p in np.arange(1, 11):
    q0, q1, obj, status = private_recs(p / 11)
    public_cost = p / 11
    soc_opt_cost = (p / 11) * (17 / 18)
    public_gain = public_cost - obj
    soc_opt_gap = obj - soc_opt_cost
    results.append(
        (
            f"{p}/11",
            round(q0, 4),
            round(q1, 4),
            round(obj, 4),
            round(public_cost, 4),
            round(public_gain, 4),
            round(soc_opt_cost, 4),
            round(soc_opt_gap, 4),
        )
    )
```

p	rule_q0	rule_q1	expected_cost	public_cost	public_gain	soc_opt_cost	soc_opt_gap
1/11	0.0000	0.1667	0.0859	0.0909	0.0051	0.0859	0.0000
2/11	0.0000	0.1667	0.1717	0.1818	0.0101	0.1717	0.0000
3/11	0.0000	0.1667	0.2576	0.2727	0.0152	0.2576	0.0000
4/11	0.0000	0.1667	0.3434	0.3636	0.0202	0.3434	0.0000
5/11	0.0000	0.1667	0.4293	0.4545	0.0253	0.4293	0.0000
6/11	0.0000	0.1667	0.5152	0.5455	0.0303	0.5152	0.0000
7/11	0.0000	0.2124	0.6037	0.6364	0.0327	0.6010	0.0027
8/11	0.0000	0.2500	0.6970	0.7273	0.0303	0.6869	0.0101
9/11	0.0258	0.2758	0.7940	0.8182	0.0242	0.7727	0.0213
10/11	0.0534	0.3034	0.8947	0.9091	0.0144	0.8586	0.0361

Problem 4 (Employment recommendations)

(a) *[update]*

(b) *[update]*

(c) *[update]*

(d) *[update]*